

M22 User Product Manual

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Revision Record

Date change	version	Version description
2021.3.8	V1.0	Initial version
2021.6.4	V1.1	Added PS2 communication mode
2021.6.30	V1.2	Added image mirror mode, inverted reading mode, and reading ability enhancement
2021.9.8	V1.3	Added invoicing mode, added the function of prohibiting all QR codes, and added Standard 2 of 5, Plessey, ChinaPost 25, Code16K, GS1 Composite, and Han Modified some errors in the operation settings of Xin, Micro PDF417, Micro QRCode, MaxiCode, and Aztec Code
2023.2.17	V1.4	Deleted the operation settings of ChinaPost 25, Micro PDF417, Han Xin, and GS1 Composite codes, and corrected some errors
2024.2.18	V1.5	Added tables and descriptions for control characters
2024.3.20	V1.6	Added the function of intercepting data segments
2024.5.21	V1.7	Delete hidden data segment function, delete PS2 keyboard port mode, delete mirror function setting code, delete enhanced identification setting code, delete enhanced identification setting, delete CodaBar Mod10 checksum, delete Standard 2 of 5, ChinaPost 25, Hanxin, Code16K code setting delete, delete QR and DM,MaxiCode and Aztec Code's ECI output settings, added Hebrew, Russian, Greek, and Thai keyboard languages, added volume level settings, added URL code blocking function, and added GS AI function (integrated with the original GSAI).
2024.7.8	V1.8	Added support for Brazilian banco codes, added UPC-A, UPC-E output system bit settings, added GS1 application identifiers, and added GS1 character substitution functions
2024.9.4	V1.81	Add ISSN, ISBN settings
2025.4.29	V1.82	Added input data encoding format and added HID-KBW compatibility mode
2025.6.12	V1.83	Added the function of using serial port commands, added the function of character substitution, added the function of command trigger mode, added the function of Read Fail information, added the function of adding AIM ID, added the

		function of output protocol for data segments, added the function of modifying the Code ID of CODE32, added the function of modifying the Code ID of ChinaPost 25, added the function of modifying the ending of two CRLF, added the function of adding the function of modifying the end of the character as CRDownArrow, and added the function of adding the function of modifying the ending character as CRDownArrow, and added the function of adding the AIM ID list. Added a list of commonly used serial port commands and added a vibration switch
2025.8.27	V1.84	Fix some errors

directory

One	Product Description.....	8
1.	Product introduction	8
2.	Product features.....	8
3.	Product parameters	8
4.	Interface description	10
Two	Setup code configuration instructions.....	11
1	Operation settings.....	11
1.1	Use the setting code.....	11
1.2	Factory reset	11
1.3	Get the firmware version and library version.....	12
1.4	User default settings	12
2.	Communication settings.....	13
2.1	Communication mode selection.....	13
2.2	Serial communication interface.....	14
2.2.1	Baud rate	14
2.2.2	Verification method.....	15
2.2.3	Using serial port commands	15
2.3	HID Device Data Sending Interval.....	31
2.4	Caps Lock Settings.....	32
2.5	USB HID-KBW	33
2.5.1	USB HID-KBW Device Class	33
3.1	Button trigger mode.....	33
3.1.1	Single code reading time	34
3.2	Continuous code reading mode	34
3.2.1	Single code reading time	35
3.2.2	The length of the reading interval	35
3.2.3	Same code reading delay.....	36
3.3	Automatic sensing mode.....	37
3.3.1	Single code reading time	37
3.3.2	Reading interval duration	38
3.3.3	Same code reading delay.....	39
3.3.4	Sensitivity	39
3.4	Command Trigger Mode.....	40
3.4.1	Mode entry.....	40
3.4.2	Single code reading time	40
4.	Fill in light and positioning.....	41
4.1	Fill light	41
4.2	Positioning.....	42
5	prompt output.....	43
5.1	Silent main switch	43
5.2	Boot prompt sound	43
5.3	Decoding the prompt tone	43

5.4 Set code prompt tone.....	44
5.4 Buzzer settings	44
5.5 Volume level setting.....	45
5.6 Data input and output encoding format	46
5.6.1 Data input coding format	46
5.6.2 Data output encoding format.....	47
5.7 Virtual keyboard enable	47
5.7.1 Virtual keyboard enablement.....	47
5.8 Keyboard settings for each country.....	50
5.9 Reverse Reading Mode.....	53
5.10 Invoicing mode.....	53
5.11 URL blocking.....	54
5.12 GS1 AI Settings	54
5.13 Read File Information	56
5.13.1 RF Information Settings.....	56
5.13.2 Modifying RF Information	56
5.14 Reading success vibration prompts.....	57
6. Data editing.....	57
6.1 Prefix.....	57
6.2 Suffix	58
6.3 CODE ID	58
6.3.1 Add CODE ID.....	58
6.3.2 Modify CODE ID.....	59
6.4 AIM ID	61
6.5 Tail.....	61
6.6 Data segments.....	62
6.6.1 Data segment interception.....	62
6.6.2 Character substitution	64
6.6.3 Output Protocol.....	66
7 Yard setting.....	66
7.1 Global quick operation	66
7.1.1 Global operation.....	66
7.1.2 Global operation.....	67
7.2 One-dimensional barcode operation	67
7.2.1 EAN13.....	67
7.2.2 ISSN	69
7.2.3 ISBN	69
7.2.4 EAN 8.....	70
7.2.5 UPC-A	71
7.2.6 UPC-E0.....	73
7.2.7 UPC-E1.....	75
7.2.8 Code128	76
7.2.9 Code39	77

7.2.10 Code93	78
7.2.11 Interleaved 2 of 5	79
7.2.12 Industrial 25	80
7.2.13 Matrix 2 of 5.....	81
7.2.14 Code11	82
7.2.15 CodaBar	83
7.2.16 MSI	84
7.2.17 RSS-14	85
7.2.18 RSS- Ltd.....	86
7.2.19 RSS- Exp.....	86
7.2.20 Plessey.....	87
7.3 QR code operation	88
7.3.1 QR Code.....	88
7.3.2 Data Matrix(DM)	88
7.3.3 PDF417	88
7.3.4 Micro PDF417	88
7.3.5 Micro QRCode	89
7.3.6 MaxiCode	89
7.3.7 Aztec Code.....	89
8 Save and cancel	90
appendix.....	90
Appendix A: Data Code.....	90
Appendix B: Default Settings Table	92
Appendix C: Code ID List	97
Appendix D: ASCII Code Table	98
Appendix E: AIM ID List	102
Appendix F: Common Serial Commands	102

一、 Product Description:

Thank you for using the products provided by Siyuanchuang. Carefully reading this document can help you understand the functions and features of this product, as well as quickly grasp the use and installation methods of the product.

The Company shall not be liable for property damage or personal injury caused by the user's abnormal operation. Users are requested to develop the corresponding product according to the technical specifications and reference designs in the manual. At the same time, pay attention to the general safety matters that should be paid attention to when using mobile products. The Company reserves the right to modify the content of this manual according to the needs of technological development before making a statement.

1. Product introduction

M22 is an image-based two-dimensional scanning module with excellent performance and high integration, which applies the country's leading image intelligent recognition algorithm, can read all kinds of one-dimensional barcodes, two-dimensional codes, has strong reading ability, can adapt to various complex environments, and can read barcodes in strong light or dark environments.

2. Product features

- Strong literacy skills
- Stable performance
- Strong compatibility

3. Product parameters

Code scanning performance	Image sensor	640 (horizontal) x 480 (vertical) CMOS
	light source	white light
	Horizontal perspective	31°
	Reading and coding system	2D: QR Code、PDF417、DataMatrix (ECC200)、Code16K、Micro QRCode、MaxiCode、Aztec Code 1D: EAN13、EAN8、UPC-A、UPC-E0、UPC-E1、Code128、Code39、Code93、CodaBar、Interleaved 2 of 5、Industrial25、Matrix 2 of 5、Code11、MSI、RSS-14、RSS-14 Limited、RSS-14 Expanded、GS1 Databar(RSS)、Standard 2 of 5、Plessey

	Deep reading field	EAN-13 4.0~20.0cm(13mil) Code128 4.5~25.0cm(15mil) QR Code 4.0~18cm(15mil)
	Reading angle	Rotate 360° , tilt $\pm 60^\circ$, deflect $\pm 60^\circ$
	Reading accuracy	1D code $\geq 5\text{mil}$ QR code $\geq 10\text{mil}$
	Symbol contrast	$\geq 25\%$
Physical parameters	weight	< 7g
	Appearance dimensions	21.4mm x 12.5mm x 11.8mm (L x W x H)
	interface	TTL-232 / USB 2.0 Full Speed
	voltage	DC +3.3V $\pm 5\%$
	current	Operating current 120mA Standby current 1mA
Environmental parameters	Operating temperature	-20°C~60°C
	Storage temperature	-40°C~70°C
	Operating humidity	5%RH~95%RH(non-condensing)
reliability	Resistant to mechanical shocks	Vibration: 5 - 200 Hz, 10 Grms, 3 axes, 1.5 hours per axis. Drop: Withstands 1.2 m (4 ft) drop onto concrete.
	Thermal shock resistant	Minimum temperature -20°C (-4°F). Maximum temperature 60°C (140°F). Cycle times: 30 minutes High temperature; 30 minutes of low temperature Cycle 24

4. Interface description

PIN	Input/output	definition	illustrate
PIN1	-	TEST1	Internal test pins External suspension is recommended
PIN2	power supply	VCC	Input +3.3V
PIN3	earth	GND	-
PIN4	input	RX	The module's TTL serial port receives
PIN5	output	TX	The module's TTL serial port sends
PIN6	input	D-	USB DN signal
PIN7	output	D+	USB DP signal
PIN8	earth	GND	-
PIN9	output	BEEP	Passive buzzer outputs signal When idle, it is low
PIN10	output	DLED	Decode the success indicator light When idle, it is low
PIN11	-	TEST2	Internal test pins External suspension is recommended
PIN12	input	TRIG	Scan the code to trigger the signal Low levels are effective

二、 Setup code configuration instructions

1 Operation settings

1.1 Use the setting code

Read "Turn on the setting code", and the function of the reading module can be configured (set code function). After the function is enabled, the module can be modified by reading one or more setting codes. After reading the "Close Setting Code", the reading module will exit the setting state.



**Turn on the setting code



Turn off the setcode

Set the code content output settings



**Do not output the content of the setting code



Output the setcode content

Note: The option marked (**) in the setting code indicates that this is the default function or parameter.

1.2 Factory reset

After reading this setting code, the current parameter settings will be lost, and the default values at the factory will be restored. The factory default parameters and functions are detailed in Appendix B



Factory reset

1.3 Get the firmware version and library version

Read the following setting code to query the current firmware version and library version



Get the firmware version



Get the library version

1.4 User default settings

In addition to factory reset, users can save frequently used settings as user defaults. By reading "Save current settings as user default configuration", the current device configuration can be saved as the user's default settings so that quick settings can be made when needed.

By reading "Revert to User Default Configuration", you can restore the default settings saved by the user.



Save the current settings as the default configuration for
the user



Revert to the user's default configuration

2. Communication settings

The M22 Literacy Module provides a TTL-232 serial communication interface and a USB interface (optional function) for communication connection with the host. Through the communication interface, it can receive reading data.

2.1 Communication mode selection

By default, USB-HID mode is used for communication from the factory. Users can switch between communication port output modes (TTL serial mode/virtual serial/USB-HID mode) through the code scanning settings. When users need to output USB and serial ports at the same time, they can read the setting code and select HID + TTL serial port mode. When the user switches communication modes, they need to wait for the device to be initialized before performing related operations.

Note: When the module is set to a USB virtual serial port and communicates with the host through this port, the host needs to install the corresponding driver.



**USB HID mode



Virtual serial port mode



TTL serial mode



HID+TTL serial port mode

2.2 Serial communication interface

Serial communication interface is a common way to connect the reading module with the host device. When using serial communication interfaces, the communication parameters between the reading module and the host device must be completely matched to ensure smooth communication and correct content.

The serial communication interface provided by the Reader Module is a TTL level signal. The TTL serial port can be used to connect to most application architectures, but when RS-232 is required, an external conversion circuit is required.

The default serial communication parameters of the reading module are as follows, which can be modified by reading the setting code if they are inconsistent with the host device.

parameter	default
Serial communication type	Standard TTL-232
Baud Rate	9600
Parity Type	None
Data Bits	8
Stop Bits	1

2.2.1 Baud rate

The unit of Baud Rate is bits per second (bps), and the following setting code can be read to select the configuration parameters.



Serial port baud rate - 1200bps



Serial port baud rate -2400bps



Serial port baud rate - 4800bps



** Serial baud rate -9600bps



Serial baud rate - 19200bps



Serial port baud rate -38400bps



Serial port baud rate - 57600bps



Serial port baud rate - 115200bps

2.2.2 Verification method

There are three optional verification methods:



** Serial check bit - none



Serial check bit - odd check



Serial check bit-even check

2.2.3 Using serial port commands

Users can send serial port commands from the host to set the reading engine. The communication parameters between the reading engine and the host device must be completely matched

Normal communication can be achieved by matching time. The default serial communication parameters of the reading engine are: **baud rate 9600bps, no check, 8 bits of data, 1 bit of stop, and no flow control.**

2.2.3.1 Read the sign bit operation

For read operations on device flag bits, up to 256 bytes of flag bits can be read at a time.

Command Format:

Format: {Head1} {Types} {Lens} {Address} {Datas}

{CRC} where Head1 : 0x7E 0x00 (2 bytes);

Types : 0x07 (1 byte);

Lens : 0x01 (1 byte);

Address: 0x0000~0x00FF (2 bytes), which indicates the starting address of the flag bit to be read;

Datas: 0x00~0xFF (1 byte), which represents the number of bytes of the flag bit to be read continuously, 0x00 represents 256 bytes;

CRC: CRC_CCITT checkis (2 bytes). The range of calculations: Types, Lens, Address, Datas, the calculation method is CRC_CCITT, and the feature polynomial: $X^{16}+X^{12}+X^5+1$, that is, the polynomial coefficient is 0x1021, the initial value is all 0, and the highest bit is calculated first for a single byte, and there is no need to reverse it, and it is output directly. The reference code for C is as follows:

```
unsigned int crc_cal_by_bit(unsigned char* ptr, unsigned int len)
{
    unsigned int crc = 0;
    while(len-- != 0)
    {
        for(unsigned char i = 0x80; i != 0; i /= 2)
        {
            crc *= 2;

            if((crc&0x10000) != 0) // After multiplying the previous CRC by 2, divide by 0x1021 if the first
digit is 1
                crc ^= 0x1021;
        }
    }
}
```



```

        if((*ptr&i) != 0) // If current data bit is 1, XOR with CRC polynomial

            crc ^= 0x1021;

    } // for(unsigned char i = 0x80; i != 0; i /= 2)

    ptr++;

} // while(len-- != 0)

return crc;

} // unsigned int crc_cal_by_bit(unsigned char* ptr, unsigned int len)

```

Note: When the user does not need the CRC verification function, they can fill in the 0xAB 0xCD in the CRC byte without verification.

Return: {Head2} {Types} {Lens} {Datas} {CRC}

1) Read successfully and return to read data

Head2 :0x02 0x00(2 bytes);

Types :0x00 means read success (1 byte);

Lens: represents the number of bytes (1 byte) of uploaded data, 0x00 represents 256;

Datas: 0x00~0xFF, which represents the data read up (1 byte);

CRC: CRC_CCITT checksis (2 bytes). The scope of calculation: Types, Lens, Datas, the calculation method is CRC_CCITT, and the feature polynomial: $X^{16}+X^{12}+X^5+1$, that is, the polynomial coefficient is 0x1021, the initial value is all 0, and the highest bit is calculated first for a single byte, and there is no need to reverse it, and it is output directly (the reference code is the same as above).

2) CRC verification failed

No response to orders

3) Unknown command response

No response to orders

Example:

Read 1 address with address 0x000A in the flag bit :

1) Read successfully and return data, returned data as 0x3E

Input: 0x7E 0x00 0x07 0x01 0x00 0x0A 0x01 0xEE 0x8A

Return: 0x02 0x00 0x00 0x01 0x3E 0xE4 0xAC

2) CRC error issued

Input: 0x7E 0x00 0x07 0x01 0x00 0x0A 0x01 0x11 0x22

Return: None

3) When the length of the sent instruction is not enough or the waiting time after sending the 0x7e 0x00 exceeds 400ms, it is treated as an unknown command

Input: 0x7E 0x00 0x07 0x01 0x00 0x0A 0x01

Return: None

2.2.3.2 Write the flag bit operation

For a write operation on a device flag bit, a maximum of 256 bytes of flag bits can be written at a time.

If you need to keep it after a power outage, you need to save the flag bit to the internal flash

(see 2.2.3.3).

Command Format:

Input: **{Head1}{Types}{Lens}{Address}{Datas}{CRC}**

Head1 :0x7E 0x00(2 bytes); Types :

0x08 (1 byte);

Lens: 0x00~0xFF (1 byte), which represents the number of bytes in the Datas field in the command, and also indicates that it is to be continuous

the number of write operations, while 0x00 means there are 256 bytes;

Address :0x0000~0xFFFF(2 bytes), which represents the starting address of the flag bit to be written;

Datas: 0x00~0xFF (1~256 bytes), which represents the data written to the flag bit, and when configuring multiple flag bits, you must press

populate the data fields in the order of address from lowest to highest;

CRC: CRC_CCITT checksis (2 bytes). The range of calculations: Types, Lens, Address, Datas, the calculation method is CRC_CCITT, and the feature polynomial: $X^{16}+X^{12}+X^5+1$, that is, the polynomial coefficient is 0x1021, the initial value is all 0, and the highest bit is calculated first for a single byte, and there is no need to reverse it, and it is directly output. The reference code for C is as follows:

```
unsigned int crc_cal_by_bit(unsigned char* ptr, unsigned int len)
{
    unsigned int crc = 0;
    while(len-- != 0)
    {
        for(unsigned char i = 0x80; i != 0; i /= 2)
```

```

{
    crc *= 2;

    if((crc&0x10000) !=0) // After multiplying the previous CRC by 2, divide by 0x11021 if
the first digit is 1

    crc ^= 0x11021;

    if((*ptr&i) != 0) // If current data bit is 1, XOR with CRC polynomial

        crc ^= 0x1021;

} // for(unsigned char i = 0x80; i != 0; i /= 2)

ptr++;

} // while(len-- != 0)

return crc;

} // unsigned int crc_cal_data_bit(unsigned short data, unsigned int len)

```

Note: When the user does not need the CRC verification function, they can fill in the 0xAB 0xCD in the CRC byte without verification.

Return: {Head2} {Types} {Lens} {Ddatas} {CRC}

1) Write successfully

Head2 :0x02 0x00(2 bytes);

Types :0x00 indicates successful write (1 byte);

Lens : 0x01 (1 byte);

Dates: 0x00 (1 byte);

CRC :CRC_CCITT ???(0x33 0x31) (2 bytes) ±

2) CRC verification failed

No response to orders

3) Unknown command response

No response to orders

Example:

Write 0x3E: to the flag bit with the address 0x000A

1) Setup successful

Input:0x7E 0x00 0x08 0x01 0x00 0x0A 0x3E 0x4C 0xCF

Return: 0x02 0x00 0x00 0x01 0x00 0x33 0x31

2) CRC error issued

Input:0x7E 0x00 0x08 0x01 0x00 0x0A 0x3E 0x11 0x22

Return: None

3) When the length of the sent instruction is not long enough or the waiting time after sending the 0x7e 0x00 exceeds 400ms, it is treated as an unknown command

Input: 0x7E 0x00 0x08 0x01 0x00 0x0A 0x3E

Return: None

2.2.3.3 Flag bits are saved to the internal Flash command

To save a list of device flag bits to internal Flash , you need to send a save command.

Note: Devices cannot save individual flag bit configurations individually, the entire list must be saved at the same time.

Command Format:

Input: **{Head1}{Types}{Lens}{Address}{Datas}{CRC}**

Head1 :0x7E 0x00(2 bytes); Types :

0x09 (1 byte); Lens :

0x01 (1 byte);

Address : 0x0000 (2 bytes);

Dates: 0x00 (1 byte);

CRC :CRC_CCITT ???(0xDE 0xC8)(2 bytes)±

Return: **{Head2}{Types}{Lens}{Dats}{CRC}**

1) Save successfully

Head2 :0x02 0x00(2 bytes);

Types :0x00 indicates successful write (1 byte);

Lens : 0x01 (1 byte);

Dates: 0x00 (1 byte);

CRC :CRC_CCITT ???(0x33 0x31)(2 bytes)±

2) CRC verification failed

No response to orders

3) Unknown command response

No response to orders

2.2.3.4 The flag position is restored to factory settings

To restore the contents of the device flag bit to factory settings and save it to the plug-in's EEPROM , you need to send a factory reset

Command.

Command Format:

Input: **{Head1}{Types}{Lens}{Address}{Dats}{CRC}**

Head1 :0x7E 0x00(2 bytes); Types

: 0x08 (1 byte);

Lens : 0x01 (1 byte);

Address: 0x00D9 (2 bytes);

Date: 0x50 (1 byte);

CRC :CRC_CCITT ???(81 D3)(2 bytes)±

Return: **{Head2}{Types}{Lens}{Dats}{CRC}**

1) Save successfully

Head2 :0x02 0x00(2 bytes);

Types :0x00 indicates successful write (1 byte);

Lens : 0x01 (1 byte);

Dates: 0x00 (1 byte);

CRC :CRC_CCITT ???(0x33 0x31)(2 bytes) ±

2) CRC verification failed

No response to orders

3) Unknown command response

No response to orders

Return: None

2.2.3.5 List of flag bits

Flag bit	0x0000
Data bits	function
Bit 7	1: Turn on the decoding success LED prompt 0: Turn off the decoding success LED prompt
Bit 6	1: Turn off mute 0: Start mute
Bit 5-4	00: No aim 01: Normal 10/11: Solid light
Bit 3-2	00: No lighting 01: Normal 10/11: Solid on
Bit 1-0	00: Manual mode 01: Command trigger mode 10: Continuous mode 11: Induction mode
Flag bit	0x0002
Data bits	function
Bit 7-1	retain
Bit 0	Command mode triggers the flag and automatically clears the scan after the scan is completed. 1: Trigger 0: Not triggered
Flag bit	0x0003
Data bits	function
Bit 7-2	HID Polling Interval. Interval = (Reg0x0003[7:2] + 1) ms.
Bit 1	1: Turn off the setting code 0: Turn on the setting code
Bit 0	1: Output the set code content 0: Do not output the set code content
Flag bit	0x0004
Data bits	function
Bit 7-0	The image stabilization time of the sensing mode is 0x00-0xFF: 0.0-25.5s
Flag bit	0x0005
Data bits	function
Bit 7-0	The length of the reading interval 0x00-0xFF: 0.0-25.5s
Flag bit	0x0006
Data bits	function
Bit 7-0	Single reading time

	0x00-0xFF: 0.0-25.5s
Flag bit	0x0009
Data bits	function
Bit 7-2	HID pre-release interval. Interval = (Reg0x0009 [7:2])ms
Flag bit	0x000B
Data bits	function
Bit 7-0	Reading success tone duration 0x00-0xFF: 0-255ms
Flag bit	0x000C
Data bits	function
Bit 7-2	HID Release-to-Release Interval. Interval = (Reg0x000C[7:2]) ms
Bit 1	CapsLock switch. 0: Close 1: On
Bit0	Default level in active buzzer mode 0: Buzzer idle high, busy low 1: Buzzer idle low, busy high
Flag bit	0x000D
Data bits	function
Bit 7	Invoicing mode Enable Flag 0: Disable 1: Enable
Bit 6	Virtual keyboard enable flag 0: Disable 1: Enable
Bit 5-4	Enter the data encoding format 00=GBK, 01=Reserved, 10=AUTO, 11=UTF-8.
Bit 3-2	Output data encoding format 00: GBK 01: UNICODE 10: Reserved (Do not use) 11: UTF8
Bit 1-0	00: Serial Port Output 01: USB PC Keyboard 10: Reserved 11: USB Virtual Serial Port
Flag bit	0x000E
Bit 7-3	retain
Bit 2	1: Turn on the reading success prompt 0: Turn off the reading success prompt tone
Bit 1	1: Turn off the startup tone 0: Turn on the startup tone
Bit 0	1: Enable serial port to simulate HID protocol 0: Disable serial port to simulate HID protocol
Flag bit	0x0010
Data bits	function
Bit 7-0	Induction sensitivity adjustment 0x00-0x0F: The higher the value, the higher the sensitivity, and the default value 0x05
Flag bit	0x0013
Data bits	function
Bit 7	The same barcode reading delay setting 0: Turn off the same barcode reading delay 1: Turn on the same barcode reading delay
Bit 6-0	Same barcode reading delay (unit: 100ms) 0x00: Infinite Length 0x01-0x7F: 0.1-12.7 seconds:
Flag bit	0x0023
Data bits	function
Bit 7-0	Decode Success LED Duration 8 Bits (1 for 10ms)

Flag bit	0x0028
Data bits	function
Bit 7-1	retain
Bit 0	Read the MAXICODE code 0: Prohibition of reading MAXICODE codes 1: Allowing reading of MAXICODE codes
Flag bit	0x0029
Data bits	function
Bit 7-1	retain
Bit 0	MICROPDF417 Code Reading 0: Reading MICROPDF417 code is prohibited 1: Reading MICROPDF417 code is allowed
Flag bit	0x002B, 0x002A
Data bits	function
Bit 15	retain
Bit 14-13	Parity mode: 00: No check 01: Odd check 10: Even check
Bit 12-0	0x09C4: Serial port baud rate is 1200bps 0x0271: Serial port baud rate is 4800bps 0x0139: Serial port baud rate is 9600bps 0x00D0: Serial port baud rate is 14400bps 0x009C: Serial port baud rate is 19200bps 0x004E: Serial port baud rate is 38400bps 0x0034: Serial port baud rate is 57600bps 0x001A: Serial port baud rate is 115200bps Example: 9600 baud rate: 0x002A=0x39, 0x002B=0x01
Flag bit	0x002C
Data bits	function
Bit 7-3	retain
Bit 2-1	Barcode master switch 00: Prohibit reading all barcodes: 01: Allow reading all barcodes; 11: Turn on the default readable barcode; 10: Retention;
Bit 0	retain
Flag bit	0x002E
Data bits	function
Bit 7-1	retain
Bit 0	Read EAN-13 code 0: Prohibited from reading EAN-13 codes 1: Reading EAN-13 codes is allowed
Flag bit	0x002F
Data bits	function
Bit 7-1	retain
Bit 0	Read EAN-8 codes 0: Prohibition of reading EAN-8 codes 1: Reading of EAN-8 codes is allowed
Flag bit	0x0030

Data bits	function
Bit 7-1	retain
Bit 0	Read UPC-A codes 0: Prohibition of reading UPC-A code 1: Reading of UPC-A code is allowed
Flag bit	0x0031
Data bits	function
Bit 7-1	retain
Bit 0	Read UPC-E codes 0: Reading UPC-E code is prohibited 1: Reading UPC-E code is allowed
Flag bit	0x0033
Data bits	function
Bit 7-1	retain
Bit 0	Read Code 128 0: Code 128 is prohibited 1: Code 128 is allowed
Flag bit	0x0034
Data bits	function
Bit 7-0	Code128 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0035
Data bits	function
Bit 7-0	Code128 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0036
Data bits	function
Bit 7-1	retain
Bit 0	Read Code39 0: Prohibited from reading Code39 codes 1: Allowing to read Code39
Flag bit	0x0037
Data bits	function
Bit 7-0	Code39 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0038
Data bits	function
Bit 7-0	Code39 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0039
Data bits	function
Bit 7-1	retain
Bit 0	Read Code93 0: Prohibition of reading Code93 codes 1: Reading of Code93 codes is allowed
Flag bit	0x003A
Data bits	function

Bit 7-0	Code93 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x003B
Data bits	function
Bit 7-0	Code93 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x003C
Data bits	function
Bit 7-2	retain
Bit 1	CodaBar code sends a start/end character 0: Do not send initiator/end character 1: Allow sending initiator/end character
Bit 0	Read the CodaBar code 0: Prohibit reading CodaBar codes 1: Allowing reading CodaBar codes
Flag bit	0x003D
Data bits	function
Bit 7-0	CodaBar information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x003E
Data bits	function
Bit 7-0	CodaBar information longest length setting 0x00-0xFF:0-255Byte
Flag bit	0x003F
Data bits	function
Bit 7-1	retain
Bit 0	Read QR codes 0: QR code reading is prohibited 1: QR code reading is allowed
Flag bit	0x0040
Data bits	function
Bit 7	retain
Bit 6	Read ITF6 0: No ITF6 code 1: ITF6 code is allowed Note: Interleaved 2 of 5 and enable checks must be enabled at the same time
Bit 5	Read ITF14 0: Prohibited from reading ITF 14 codes 1: Allowed to read ITF 14 codes Note: Interleaved 2 of 5 and enable checks must be enabled at the same time
Bit 4	retain
Bit 3	Transfer checksum settings 0: Does not transmit checksum 1: Transmits checksum character
Bit 2	Check the settings 0: Prohibit checks 1: Enable checks
Bit 1	retain
Bit 0	Interleaved 2 of 5 Reading 0: Disabled 1: Enabled

Flag bit	0x0041
Data bits	function
Bit 7-0	Interleaved 2 of 5 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0042
Data bits	function
Bit 7-0	Interleaved 2 of 5 Maximum Length Setting 0x00-0xFF:0-255Byte
Flag bit	0x0043
Data bits	function
Bit 7-1	retain
Bit 0	Industrial 25 Reading 0: Disabled 1: Enabled
Flag bit	0x0044
Data bits	function
Bit 7-0	Industrial 25 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0045
Data bits	function
Bit 7-0	Industrial 25 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0046
Data bits	function
Bit 7-1	retain
Bit 0	Matrix 2 of 5 Reading 0: Matrix 2 of 5 is prohibited 1: Matrix 2 of 5 is allowed
Flag bit	0x0047
Data bits	function
Bit 7-0	Matrix 2 of 5 information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0048
Data bits	function
Bit 7-0	Matrix 2 of 5 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0049
Data bits	function
Bit 7-1	retain
Bit 0	Read Code 11 0: Prohibited from reading Code 11 1: Allowed to read Code 11
Flag bit	0x004A
Data bits	function
Bit 7-0	Code 11 information minimum length setting

	0x00-0xFF:0-255Byte
Flag bit	0x004B
Data bits	function
Bit 7-0	Code 11 information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x004C
Data bits	function
Bit 7-1	retain
Bit 0	Read MSI codes 0: Prohibit reading MSI code 1: Allow reading MSI code
Flag bit	0x004D
Data bits	function
Bit 7-0	MSI information minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x004E
Data bits	function
Bit 7-0	MSI information maximum length setting 0x00-0xFF:0-255Byte
Flag bit	0x004F
Data bits	function
Bit 7-1	retain
Bit 0	Read RSS-14 codes 0: Reading RSS-14 code is prohibited 1: Reading RSS-14 code is allowed
Flag bit	0x0050
Data bits	function
Bit 7-1	retain
Bit 0	Read qualified RSS codes 0: It is forbidden to read RSS codes 1: It is allowed to read RSS-14 codes
Flag bit	0x0051
Data bits	function
Bit 7-1	retain
Bit 0	Read extended RSS codes 0: Prohibit reading extended RSS codes 1: Allow reading extended RSS codes
Flag bit	0x0052
Data bits	function
Bit 7-0	RSS message minimum length setting 0x00-0xFF:0-255Byte
Flag bit	0x0053
Data bits	function
Bit 7-0	Maximum length setting for RSS messages 0x00-0xFF:0-255Byte
Flag bit	0x0054

Data bits	function
Bit 7-1	retain
Bit 1	retain
Bit 0	Read DM codes 0: Prohibit reading DM codes 1: Allow reading DM codes
Flag bit	0x0055
Data bits	function
Bit 7-1	retain
Bit 0	Read PDF417 code 0: Prohibited from reading PDF417 code 1: Allowing reading of PDF417 code
Flag bit	0x0056
Data bits	function
Bit 7-1	retain
Bit 0	Read ISSN codes 0: It is forbidden to read the ISSN code 1: The ISSN code is allowed to be read
Flag bit	0x0057
Data bits	function
Bit 7-1	retain
Bit 0	Read ISBN codes 0: It is forbidden to read ISBN codes 1: ISBN codes are allowed to be read
Flag bit	0x005C
Data bits	function
Bit 0	Read AZTEC codes 0: Prohibited from reading AZTEC codes 1: Allowed to read AZTEC codes
Flag bit	0x005E
Data bits	function
Bit 7-2	retain
Bit 1	Decode the success tone 0: Turn off the decoding success tone 1: Turn on the decoding success tone
Bit 0	Set a success tone 0: Turn off the success tone 1: Turn on the success tone to set up
Flag bit	0x005F
Data bits	function
Bit 7-1	retain
Bit 0	Read the MICROQR code 0: Prohibit reading MICROQR codes 1: Allow reading MICROQR codes
Flag bit	0x0060
Data bits	function
Bit 7	Whether the serial/virtual serial output comes with a protocol 0: Raw data 1: With protocol
Bit 6-5	The suffix type after the terminator 00:CR(0x0D) 01:CRLF(0x0D,0x0A)

	10: TAB(0x09) 11: TWO CRLF(0x0D,0x0A,0x0D,0x0A)
Bit 4	1: Allow to add RF information 0: Prohibit the addition of RF information
Bit 3	1: Allow prefix 0: Prohibit the addition of prefixes
Bit 2	1: Allow to add Code ID 0: Prohibit the addition of Code ID
Bit 1	1: Allow adding a suffix 0: Prohibit the addition of a suffix
Bit 0	1: Allow closing characters 0: Prohibit closing characters
Flag bit	0x0061
Data bits	function
Bit 7-0	Keyboard settings for each country 00: USA 01: Czech Republic 02: France 03: Germany 04: Hungary 05: Italy 06: Japan 07: Spain 08: Turkey Q 09: Turkey F 0A: Portugal 0B: Brazil 0C: United Kingdom 0D: Netherlands 0E: Finland 0F: Denmark 10: Poland 11: Sweden 12: Belgium 13: Norway 14: Slovakia 15: Romania 16: Israel 17: Switzerland 18: Latin America 19: Russian 1A: Canada 1B: Greek 1C: Thai
Flag bit	0x00B0
Data bits	function
Bit 7-2	retain
Bit 1-0	Data character interception settings 00: Transfer all Data characters 01: Transfer only the first M Data characters 10: Only the last N Data characters are transmitted 11: N Data characters after the front M+ are not transmitted
Flag bit	0x00B1
Data bits	function
Bit 7-0	Front truncation length M 0x00-0xFF: 0-255 characters
Flag bit	0x00B2
Data bits	function
Bit 7-0	Length N is intercepted 0x00-0xFF: 0-255 characters
Flag bit	0x00D9 (only write the logo bit)
Data bits	function
Bit 7-0	Functional flag bits 0x50: Factory reset 0x55: Restore the user's default settings 0x56: The current settings are saved as the default settings for the user
Flag bit	0x00E1 (read-only flag bit)
Data bits	function
Bit 7-0	Hardware version 0x64: V1.00 0x6E: V1.10 0x78: V1.20 0x82: V1.30

	0x8C: V1.40
Flag bit	0x00E2 (read-only logo bits)
Data bits	function
Bit 7-0	Software version 0x64: V1.00 0x6E:V1.10 0x78: V1.20 0x82: V1.30 0x8C: V1.40
Flag bit	0x00E3 (read-only sign bit)
Data bits	function
Bit 7-0	Software year (2000 plus the value indicates the year) 0x17: 2023 0x18: 2024 0x19: 2025
Flag bit	0x00E4 (read-only sign bit)
Data bits	function
Bit 7-0	Software month (this value indicates the month) 0x09: September 0x0A: October 0x0B: November
Flag bit	0x00E5 (read-only flag bits)
Data bits	function
Bit 7-0	Software date (this value indicates the date) 0x09: No. 9 0x0A:10 0x0B:11

2.3 HID Device Data Sending Interval

Read the following setting code to modify the HID device data transmission interval, cycle range: 1ms~16ms



****USB data interval - 1ms**



USB data interval - 2ms



USB data interval - 4ms



USB data interval - 8ms



USB data interval - 16ms

2.4 Caps Lock Settings



***No case conversion**



Convert to uppercase



Convert to lowercase



Case swapping

2.5 USB HID-KBW

2.5.1 USB HID-KBW Device Class

You can toggle the HID KBW by scanning the following setting code. "Compatibility Mode 1" can be compatible with Sogou input method.



\$Set#Code^F2

HID KBW Chinese Input Method Compatibility Mode
is turned off

*Compatible with Mode 1



\$Set#Code^F3

HID KBW Chinese Input Method Compatibility Mode
is enabled

Compatibility Mode 2

3.1 Button trigger mode

The key trigger mode is the silent reading mode. In the button trigger mode, press the trigger button to start shooting and reading the reading module; In "Single code reading time", if the reading is successful, the reading module will output the reading content through the communication interface and stop reading. If you need to start a new reading, you need to retrigger the button. If the reading exceeds the duration of a single code reading, the shooting will be stopped.



**Push-button trigger mode

3.1.1 Single code reading time

The single reading time refers to the maximum time allowed to be taken after the reading is triggered and when the reading is not successful. In the super
After this time, it will exit the reading state. The single reading time ranges from 1s ~ infinite. The following setting code can be read once
The setting of the reading time.



Button trigger mode - single reading time - 1 second



Push-button trigger mode - single reading time - 3 seconds



Push-button trigger mode - single reading time - 5 seconds



Button trigger mode - single reading time - 8 seconds



Button trigger mode - single reading time - 10 seconds



**Push-button trigger mode - single reading time - infinite length

3.2 Continuous code reading mode

Continuous mode is a working method in which the reading module continuously cycles to shoot, read and output information. In this mode, the reading interval of 1s is entered by default after a successful read.



Continuous barcode reading mode

3.2.1 Single code reading time



Continuous barcode reading mode - single reading time
- 1 second



Continuous reading mode - single reading time - 3
seconds



** Continuous barcode reading mode - single barcode
reading time - 5 seconds



Continuous reading mode - single reading time - 8
seconds



Continuous barcode reading mode - single reading time
- 10 seconds



Continuous reading mode - single reading time - infinite
length

3.2.2 The length of the reading interval

It refers to the interval required for the next reading after successful reading. No collection and reading will be carried out during this interval. Read the following setting codes to set the reading interval time. The setting range is 0s~5s, and the default duration is 1s.



Continuous code reading mode - reading interval - no
interval



Continuous code reading mode - reading interval -
500ms



** Continuous code reading mode - reading interval - 1 second



Continuous barcode reading mode - reading interval - 2 seconds



Continuous reading mode - reading interval - 3 seconds



Continuous code reading mode - reading interval - 5 seconds

3.2.3 Same code reading delay

In order to avoid the same code being read multiple times in continuous mode, the reading module can be required to delay the setting time in this mode before allowing the same barcode to be read. The same code reading delay refers to the refusal to read the same code within a set period of time after reading a barcode. Only after the time has passed can it be read and output. By default, the same code reading delay is turned off.



The same code reading delay is turned on



**The same code reading delay is turned off

The following setting code can be read to set the same code reading delay time. Setting range: 0s~5s.

Note: You need to enable "Same Code Reading Delay" before you can set the delay duration.



Same code reading delay - unlimited delay



The reading delay of the same code is -500ms



The same code reading delay is -1 second



The same code reading delay is -2 seconds



The reading delay of the same code is -3 seconds



**The same code reading delay is -5 seconds

3.3 Automatic sensing mode

Automatic sensing mode refers to a working mode in which the reading module enters the reading mode by sensing the brightness change of the surrounding environment. When the scene changes, the reading module starts reading. After the reading successfully outputs information or the single code reading time exceeds, the reading module needs to be interacted with a certain amount of time (can be set) to re-enter the monitoring state. If the following situations do not occur, the reading module will work in a cycle as follows: If the barcode is not scanned within a single reading time, the reading module will automatically pause the reading and enter the monitoring state. In the inductive reading mode, the reading module can also start reading the code after pressing the trigger button, and continue to monitor the brightness of the surrounding environment when the code reading is successful in outputting information or releasing the trigger button.



Auto-sensing mode

3.3.1 Single code reading time



Auto-sensing mode - single reading time - 1 second



Auto-sensing mode - single reading time - 3 seconds



** Auto sensing mode - single reading time - 5 seconds



Auto-sensing mode - single reading time - 8 seconds



Auto-sensing mode - single reading time - 10 seconds



Auto-sensing mode - single reading time - infinite

3.3.2 Reading interval duration



Auto-sensing mode - reading interval - no interval



Auto-sensing mode - reading interval - 500ms



**Auto-sensing mode - reading interval - 1 second



Auto-sensing mode - reading interval - 2 seconds



Auto-sensing mode - reading interval - 3 seconds



Auto-sensing mode - reading interval - 5 seconds

3.3.3 Same code reading delay

For the same code reading delay setting, please refer to Section 3.2.3 to set the same code reading delay setting code

3.3.4 Sensitivity

Sensitivity refers to the degree of change in the detected scene in auto-sensing mode. When the reading module judges that the degree of change in the scene is met
When calculating, it will switch from monitoring state to reading state.



Auto-sensing - low sensitivity



* * Auto sensing - medium sensitivity



Auto-sensing - high sensitivity

3.4 Command Trigger Mode

3.4.1 Mode entry

In this mode, the reading engine starts reading the code when it receives a scan command sent by the host (i.e., bit0 of the flag bit 0x0002 writes "1"), and stops reading the code after the successful output of the code reading or the end of the single reading time.



Command trigger mode

Note: In command trigger mode, the serial port command that triggers the scan is: 7E 00 08 01 00 02 01 AB CD; After receiving the trigger command, the reading engine will first output a seven-byte response message and start scanning simultaneously (response message content: 02 00 00 01 00 33 31).

3.4.2 Single code reading time

In command-triggered reading mode, this parameter refers to the maximum time that allows the reading engine to continuously collect and recognize before the reading is successful. The single reading time is set in the range of 0.1~25.5 seconds, and the step size is 0.1 seconds. When set to 0 , it means that the reading time is infinite. The default duration is 5.0 seconds.



Command trigger mode - single code reading time - 1 second



Command trigger mode - single reading time - 3 seconds



* Command trigger mode - single reading time - 5 seconds



Command trigger mode - single reading time - 8 seconds



Command trigger mode - single reading time - 10
seconds



Command trigger mode - single reading time - infinite
length

4. Fill in light and positioning

4.1 Fill light

A set of LEDs on the reading module is specially equipped to provide supplementary light when shooting reading, shining the beam on the reading target, improving reading performance and adaptability in low ambient light. You can set it according to the actual usage conditions:

Fill light - light up when taking pictures: The fill light is on when shooting and turns off at other times.

Fill light - always-on: The fill light continues to emit light after the reading module is turned on.

Fill light - always off: Under no circumstances does the fill light come on.



**The fill light is lit when taking photos



The fill light is always off



The fill light is always on

4.2 Positioning

There is an auxiliary positioning device on the reading module, which projects an indicator line when shooting and reading, prompting the user to read the center of the scene image taken by the reading module.

Positioning light on when taking photos: The positioning light turns on when shooting and turns off at other times.

Positioning light is always on: The positioning light continues to shine after the reading module is turned on.

Positioning light is always off: Under no circumstances does the positioning light come on.



** The positioning light lights up when taking pictures



The positioning light is always on



The locating light is always off

5 prompt output

5.1 Silent main switch

Read the following setting codes to turn on/off all tones



** Buzzer silent off



The buzzer is muted on

5.2 Boot prompt sound

When the reading module is successfully powered on, it can turn on/off the power-on prompt sound according to the setting requirements



Turn on the power on tone



Turn off the power on tone

5.3 Decoding the prompt tone

Read the following setting codes to turn on/off the decoding success prompt tone



** Turn on decoding tone



Turn off the decoding tone

5.4 Set code prompt tone

Read the following setting codes to turn on/off the setting code prompt tone



** Turn on the setting code prompt tone



Turn off the setup code tone

Users can set the duration of the decoding prompt tone by reading the following setting code.



The duration of the decoding prompt tone is -30ms



**Decoding the prompt tone duration -60ms



The duration of the decoding prompt tone is -90ms



The duration of the decoding prompt tone is -120ms

5.5 Buzzer settings

Passive buzzer setup

Reading the following setting code sets the buzzer to passive and sets the drive frequency of the passive buzzer



**** Passive buzzer**

Passive buzzer - low-frequency tone



* * Passive buzzer - midrange tone



Passive buzzer - high-frequency tone

Active buzzer settings

Reading the following setting code sets the buzzer to active and sets the operating level with the buzzer. Read "High Level Effective", the buzzer is set to low level effective when idle and high level effective when working; Read "Low Active", the buzzer is set to high active when idle and low active when operating.



Active buzzer

**** Active buzzer - high level effective**

Active buzzer - low level effective

5.6 Volume level setting

Users can set the volume level of the reading module through the following setting code



Volume level 0



Volume level 1



Volume level 2



Volume level 3



Volume level 4



Volume level 5



Volume level 6



** Volume level 7

5.7 Data input and output encoding format

5.7.1 Data input coding format

In order for the scanning platform to read Chinese barcodes in various encoding formats, it can be set by scanning the "Input Data Encoding Format".

Note: "Input Data Encoding Format AUTO" is not compatible with "Input Data Encoding Format BIG5".



\$Set#Code^1C

Enter the data encoding format GBK



\$Set#Code^1A

Input data encoding format UTF8



\$Set#Code^1D

*Enter the data encoding format AUTO



\$Set#Code^1B

Input data encoding format BIG5

5.7.2 Data output encoding format

Users can set the output format of the reading module through the following setting code, so that the host can output Chinese data according to the specified encoding format.

Note: GBK format is used for notepads, and UNICODE format is used for Word and common chat tool input boxes.



**Data output encoding format UTF-8



The data output is encoded in Unicode



Data output encoding format GBK

5.8 Virtual keyboard enable

5.8.1 Virtual keyboard enablement

In order to adapt to the application environment in more regions, the output settings of the standard/virtual keyboard can be set by reading the following setting code. But it will be damaged

Loss of certain output efficiency. Note that when using a virtual keyboard, you must ensure that the numeric keys of the keypad are enabled.



** Standard keyboard mode



Virtual keyboard mode

1. Virtual keyboard output method

In order to be able to adapt to different application scenarios, the virtual keyboard has 2 different output methods for control characters less than 0x20, users
You can switch by scanning the setting code below.



**Control character-Ctrl mode



Control Character - Alt Mode

2. Control character sending

ASCII characters between 0x00~0x1F can be escaped into a control function key. When the virtual keyboard is enabled (the other HID keyboards of the module are set to default values), the input operation of the control function key is as follows: (For the specific relationship between ASCII values and control function keys, please refer to the control character correspondence table)

(1) The virtual keyboard Ctrl mode is turned on

The reading data is "A<HT>F (HT is an invisible character, not displayed on the terminal software)" (hexadecimal values are 0x41/0x09/0x46), and the virtual keyboard operation of the scanning module is as follows:

Enter "A" - press button A;

Enter "Ctrl I" - since the data of the 0x09 corresponds to the control function key "I", the virtual keyboard will hold down Ctrl, then press the I key, and finally release the Ctrl key and the I key at the same time;

Type "F" - press the key F.

Since "Ctrl I" corresponds to the ability to convert characters to italics in some word processing software, you may see the above when you complete the above operation

The normal character "A" and the italic "F".

Currently, the virtual keyboard Ctrl mode controls character output only supports US keyboard layouts.

(2) Virtual keyboard Alt mode

If the virtual keyboard is enabled and set to Alt mode, the output control character operation is: ALT+"Character corresponds to ASCII code base 10 value". For example, for the "<HT>" character, the virtual keyboard operation of the scan module is as follows:

Type "Alt 0 9" - the virtual keyboard will hold down Alt, then press "0" and "9" on the number pad, and finally release

Alt.

When the standard keyboard output, the control character output function is turned off, and ASCII characters less than 0x20 will output the corresponding key-value function.

(For the corresponding functions, please refer to the control character correspondence table.)

ASCII Function	ASCII Value (HEX)	Control character output off	Control the character output Ctrl mod	CTRL+X function
NOBODY	00	Null	Ctrl+@	

SOH	01	KeypadEnter	Ctrl+A	Select all
STX	02	CapsLock	Ctrl+B	Bold
ETX	03	OLD	Ctrl+C	Copy
EOT	04	Null	Ctrl+D	Bookmark
ENQ	05	CTRL	Ctrl+E	Center
ACK	06	Null	Ctrl+F	Find
BEL	07	Enter	Ctrl+G	
BS	08	LeftArrow	Ctrl+H	History
HT	09	Horizontal Tab	Ctrl+I	Italic
LF	0A	DownArrow	Ctrl+J	Justify
VT	0B	Vertical Tab	Ctrl+K	hyperlink
FF	0C	Delete	Ctrl+L	list,left align
CR	0D	Enter	Ctrl+M	
SO	0E	Insert	Ctrl+N	New
YES	0F	Esc	Ctrl+O	Open
According to	10	F11	Ctrl+P	Print
DC1	11	Home	Ctrl+Q	Quit
DC2	12	PrintScreen	Ctrl+R	
DC3	13	Backspace	Ctrl+S	Save
DC4	14	tab+shift	Ctrl+T	
TO	15	F12	Ctrl+U	
AMAZED	16	F1	Ctrl+V	Paste
ETB	17	F2	Ctrl+W	
CAN	18	F3	Ctrl+X	
IN	19	F4	Ctrl+Y	
SUB	1A	F5	Ctrl+Z	
ESC	1B	F6	Ctrl+[	
FS	1C	F7	Ctrl+\	

GS	1D	F8	Ctrl+]	
RS	1E	F9	Ctrl+6	
US	1F	F10	Ctrl+-	

5.9 Keyboard settings for each country

In order to adapt to each country, the device can be set to the corresponding "keyboard" for each country via the following setting code



** United States



Belgium



Brazil



Canada



Croatia



Czech Republic



Slovakia



Denmark



Finland



France



Germany



Hungary



Italy



Latin America



Netherlands



Norway



Poland



Portugal



Serbia



Slovenia



Spain



Sweden



Swiss-French



Swiss-German



United Kingdom



Turkey-F



Turkey-Q



Japanese



Romania



Hebrew



Russia



Greece



Thailand

5.10 Reverse Reading Mode

In some special application scenarios, it is necessary to read special barcodes that are inverted in black and white. Users can enable/disable the reading function of inverted barcodes by reading the following setting codes.



**Reverse color recognition is prohibited



Allows reverse color recognition

5.11 Invoicing mode

In order to facilitate the use of this module in the invoicing system, users can configure the invoicing mode by reading the setting code to realize the invoice code format conversion and output.



* * Invoicing mode enabled



The invoicing mode is inactive

5.12 URL blocking

Block URLs starting with "http://" or "https://".



URL blocking enabled



** URL blocking prohibition

5.13 GS1 AI Settings

Support GS1 rules to display AI fields



Turn on GS AI



**Turn off GS AI

GS1 parentheses enable character conversion

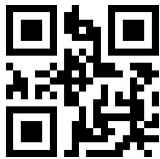


GS1 parenthetical character conversion enablement



**GS1 parenthetical character conversion is prohibited

GS1 bracket conversion mode selection



GS1 parentheses do not convert



GS1 parentheses are converted to "\"|\"/>



GS1 parentheses are converted to "\"\"/>



GS1 parentheses are converted to "\"^"/>



GS1 parentheses are converted to "\"<GS>"/>



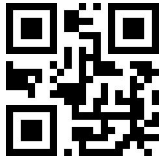
GS1 parentheses are converted to "\"(GS)"/>



GS1 parentheses are converted to "\"'GS'"/>



GS1 parentheses are converted to 'GS'



GS1 parentheses are converted to "\"?"/>



GS1 Parentheses are converted to "\"*"/>



GS1 parentheses are converted to "\"[GS]"/>



GS1 parentheses are converted to 0x1D

5.14 Read File Information

5.14.1 RF Information Settings

RF (Read Fail) information setting can enable the reading engine to output some user-defined information when the code reading is unsuccessful in certain modes, and the user or program can make corresponding adjustments or operations after detecting this information.



Send RF information



** No RF messages are sent

5.14.2 Modifying RF Information

Scan the "Modify RF Information" setting code, and combine the scanning data setting code, users can modify the RF information content for each RF character

Using two hexadecimal values for representation, RF allows up to 15 characters, and the hexadecimal conversion table for character values can be referenced (Appendix D: ASCII code list).



Modify RF information

Example: Modify the user-defined RF information to "FAIL"

1. Look up the character table (Appendix D: ASCII code list) and get the hexadecimal values of the four characters "FAIL": "46", "41", "49" 、 "4C" ；
2. Confirm whether the setting code is enabled, if not, scan the "Enable Setting Code" setting code (see Section 1.1);
3. Scan the "Modify RF Information" setting code;
4. Scan the data setting code (Appendix A: Data code) "4", "6", "4", "1", "4", "9" 、 "4" 、 "C" ；
5. Scan the "Save" setting code (8: Save or Cancel).

5.15 Reading success vibration prompts



Reading success vibration opens



**Reading successfully shakes off

6. Data editing

In practical applications, in order to facilitate data differentiation and processing, it is sometimes necessary to edit the reading data before output.

Data redaction includes:

- Add Prefix (Prefix)
- Add Suffix (Suffix)
- Decode data segment intercepts
- Output barcode CODE ID
- Decoding Failure Feature Output Information (RF Information)
- Add Tail

Processed output data format:

【Prefix】【CODE ID】【Data】【Suffix】【Tail】

6.1 Prefix

The prefix is a string of characters that is customized by the user before the decoded data. Users can add prefixes by reading the following setting codes

Additions and modifications.



Allow the addition of prefixes



** Prefixes: none

Read the following setting code, and with the "Data Code" setting code and "Save" setting code, users can modify the prefix content.



Modify the prefix

Note: Prefixes allow up to 15 characters, and for each prefix character, two hexadecimal values are used. The hexadecimal conversion table for character values is shown in Appendix D

6.2 Suffix

The suffix is a user-defined string after the decoded data. Users can read the following setting code for suffixes Add and modify.



Allow the addition of suffixes



** Suffixes: none

Read the following setting code, and with the "Data Code" setting code and the "Save" setting code, users can modify the suffix content.



Modify the suffix

Note: The suffix allows up to 15 characters, and for each suffix character, two hexadecimal values are used. The hexadecimal conversion table for character values is shown in Appendix D

6.3 CODE ID

6.3.1 ADD CODE ID

CODE ID uses one character. Users can add CODE IDs by reading the following setting codes to identify different barcode types.



Open the CODE ID



**Close the CODE ID

Read the following setting codes to restore the default CODE ID value of the barcode. The default list can be found

in Appendix C



CODE ID reverts to default

6.3.2 MODIFY CODE ID

Users can modify the CODE ID corresponding to each barcode by reading the following setting codes.



Modify the CODE ID of EAN13



Modify the CODE ID of EAN8



Modify the CODE ID of the UPC-A



Modify the CODE ID of UPC-E0



Modify the CODE ID of UPC-E1



Modify the CODE ID of Code128



Modify the CODE ID of Code39



Modify the CODE ID of Code93



Modify the CODE ID of Codabar



Modify Interleaved 2 of 5 的 CODE ID



Modify the CODE ID of Industrial 25



Modify the CODE ID of Matrix 2 of 5



Modify the CODE ID of Code11



Modify the CODE ID of the MSI



Modify the CODE ID of the RSS



Modify the CODE ID of the qualified RSS



Modify the CODE ID of the extended RSS



Change the CODE ID of the QR CODE



Modify the CODE ID of the Data Matrix



Modify the CODE ID of PDF417



Modify the Code ID of CODE32



Modify the code ID of ChinaPost 25

6.4 AIM ID

AIM is the abbreviation of Automatic Identification Manufacturers, and AIM ID defines identification codes for various standard barcodes (AIM IDs cannot be customized by users), as defined in (Appendix. E:AIM ID list). The reading engine can add this identification code before the barcode data after decoding, in the format: "]" + letter "C" + number "0", such as Code 128 has an AIM ID of "]C0".

ADD AIM ID

Users can identify different barcode types by AIM ID, and the AIM ID corresponding to each barcode type cannot be freely modified by the user, and the AIM ID is identified with three characters.



Allows the addition of AIM IDs



** Adding AIM IDs is prohibited

6.5 Tail



There is no end symbol



** End - Enter



Closing - Enter line break



Terminator -TAB



Closing - Two carriage returns



结束符-CRDownArrow

6.6 Data segments

6.6.1 Data segment interception

This function can be enabled when the user only needs to output a part of the decoded information.

We divide the decoding information [Data] into three parts:

【Start】【Center】【End】

The character length of the Start and End segments can be controlled by scanning the code.

Users can choose to output the decoding information of the corresponding position by scanning the code as follows.



*Transfer entire data segments



Transmit only the Start segment



Transmit only the End segment



Transfer only the Center segment

Modify the Start segment length M

Scan "Cut Length M before Modifying", and combine the scan data setting code to modify the length of the Start segment, the Start segment is allowed to be up to 255 characters, the pre-intercept length M is represented by a hexadecimal character, and the hexadecimal value conversion table corresponding to the length M can be referred to (Appendix D: ASCII Code List).



Truncated length M before modification

Modify the End segment length N

Scan "Modified Intercept Length N", and combine the scan data setting code to modify the length size of the End segment, the End segment is allowed to be up to 255 characters, and the sub-intercepted length N is represented by a hexadecimal character, and the hexadecimal value conversion table corresponding to the length N (see Appendix D: ASCII Code List).



Modified intercept length N

Transmit only the Start segment

Example: When the decoding message is "1234567890123ABC", output the first thirteen bytes "1234567890123"

1. Look up the character table (Appendix D: ASCII code list) and get the hexadecimal character corresponding to the decimal data "13" is "0D";
2. Check whether the setting code is turned on, and if not, scan the "Enable Setup Code" setting code (see 1.). 1 chapter);
3. Scan the "Transfer Start Segments Only" setup code
4. Scan the "Cut length M before modification" setting code;
5. Scan the data setting code (Appendix A: Data code) "0" and then "D" sequentially;
6. Scan the "Save" setting code (8: Save or Cancel);

Transmit only the End segment

Example: When the decoding information is "1234567890123ABC", the last three bytes "ABC" are output

1. Look up the character table (Appendix D: ASCII code list) and get the hexadecimal character corresponding to the decimal data "3" is "03";
2. Check whether the setting code is turned on, and if not, scan the "Enable Setup Code" setting code (see 1.). 1 chapter);
3. Scan the "Transfer End Segment Only" setup code
4. Scan the "Modified Intercept Length N" setting code;

5. Scan the data setting code (Appendix A: Data code) "0", then "3";
6. Scan the "Save" setting code (8: Save or Cancel);

Transfer only the Center segment

Example: When the decoding message is "12345678900123ABC", output the middle four bytes "0123"

1. Looking up the character table (Appendix D: ASCII code list), the hexadecimal characters corresponding to the decimal data "10" and "3" are "0A" and "03", respectively;
2. Check whether the setting code is turned on, and if not, scan the "Enable Setup Code" setting code (see 1.). 1 chapter);
3. Scan the Transfer Center Segments Only setting code
4. Scan the "Modified Intercept Length N" setting code;
5. Scan the data setting code (Appendix A: Data code) "0", then "3";
6. Scan the "Save" setting code (8: Save or Cancel);
7. Scan the "Cut length M before modification" setting code;
8. Scan the data setting code (Appendix A: Data code) "0" and then "A" in turn;
9. Scan the "Save" setting code (8: Save or Cancel);

6.6.2 Character substitution

Note: After the enable character replacement function is enabled, make sure that the modified character to be replaced and the modified character to be modified are set in place.



Character substitution is prohibited



Enable character replacement

Modify the replaced character

By scanning the "Modify Replaced Character" setting code and combining the scan data setting code, the user can set the replaced character to the maximum length of the replaced character to be 15 bytes. Each character is represented by a hexadecimal value, and the hexadecimal conversion

table for the character values can be referred to (Appendix D: List of ASCII codes).



Modify the replaced character

Example: Set the replaced character to “%”

1. Check the character table (Appendix D: ASCII code list) and get the hexadecimal value corresponding to the character “%” is “25”;
2. Confirm whether the setting code is enabled, if not, scan the “Enable Setting Code” setting code (see Section 1.1);
3. Scan the “Modify Replaced Characters” setting code;
4. Scan the data setting code (Appendix A: Data code) “2” and then “5” in turn;
5. Scan the “Save” setting code (8: Save or Cancel).

Modify the replaced characters

By scanning the “Modify Replaced Characters” setting code and combining the scanning data setting codes, users can set the replaced characters, and the maximum length of the replaced characters is 15 bytes. Each character is represented by a hexadecimal value, and the hexadecimal conversion table for the character values can be referenced (Appendix D: List of ASCII codes).



Modify the replaced characters

Example: Set the replacement character to “Enter CR”

1. Check the character table (Appendix D: ASCII code list) and get the hexadecimal value

corresponding to the character "CR" is "0D";

2. Confirm whether the setting code is enabled, if not, scan the "Enable Setting Code" setting code (see Section 1.1);
3. Scan the "Modify Replaced Characters" setting code;
4. Scan the data setting code (Appendix A: Data code) "0" and then "D" in turn;
5. Scan the "Save" setting code (8: Save or Cancel).

6.6.3 Output Protocol

By scanning the following setting code, you can modify the output format of the decoding result in serial/virtual serial mode. Among them, the output format with protocol is: <03>< length >< decoded data >.



With agreement



*raw data

7 Yard setting

7.1 Global quick operation

7.1.1 Global operation

Users can read the following setting codes to perform a global view of all supported code systems

Allow/prohibit reading and enable silent reading classes

type of operation. After prohibiting the reading of all types of codes, only the reading of setting codes is allowed.



All types are allowed



It is forbidden to read all types



** Opens the Silent Reading type



All QR codes are prohibited

7.1.2 Global operation

By reading the following setting codes, you can enable, prohibit, and disable the ability to enhance the ability to read barcodes. After the literacy ability is strengthened, the ability to read special codes such as defaced barcodes and QR code surfaces can be improved. Strengthening literacy prohibition will increase decoding speed.



**Enhanced identification is prohibited



Enable reinforcement of recognition

7.2 One-dimensional barcode operation

7.2.1 EAN13

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



**EAN13 check bit output is allowed



Prohibit EAN13 check bit output

Permission and prohibition of reading function

Users can allow or disable EAN13 barcode reading functions by reading the following setting codes.



**Allow reading EAN13 barcode



Prohibition of EAN13 literacy

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or disable the EAN13 additional code forced output function.



EAN13 mandatory output of additional codes



**EAN13 does not require an additional code to be output

Users can read the following setting code to enable and prohibit EAN13 additional code enablement related configurations.



EAN13-2-bit Add-on Code Enable



**EAN13-2 digit add-on code inhibition



EAN13-5-bit Additional Code Enable



**EAN13-5 bit additional code is incapable

7.2.2 ISSN

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the ISSN barcode reading function.



ISSN literacy is allowed



** Prohibition of reading ISSN

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or prohibit the forced output of ISSN additional codes.



ISSN forces the output of additional codes



**ISSN does not require the output of additional codes

Users can read the following setting code to enable and prohibit ISSN additional code configurations.



**ISSN 2-digit appendix enablement



ISSN 2-digit appendix is prohibited

7.2.3 ISBN

Permission and prohibition of reading function

Users can allow or disable the ISBN barcode reading function by reading the following setting codes.



ISBN is allowed



** Prohibited from reading ISBNs

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or prohibit the forced output of ISSN additional codes.



ISBN mandatory output of additional codes



**ISBN does not require an additional code to be output

Users can read the following setting code to enable and prohibit ISBN additional code configurations.



**ISBN 5-digit appendix enablement



ISBN 5-digit additional code is prohibited

7.2.4 EAN 8

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



**Allows EAN8 check bit output



Prohibit EAN8 check bit output

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the EAN8 barcode reading function.



**Allow reading EAN8 barcode



Prohibiting EAN8 literacy

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or disallow the forced output function of EAN8 additional codes.



EAN8 forces the output of additional codes



**EAN8 does not require additional codes to be output

Users can read the following setting code to enable and prohibit EAN8 additional code enablement related configurations.



EAN8-2-bit Additional Code Enable



**EAN8-2 digit appendix is incapable



EAN8-5-bit Add-on Code Enable



**EAN8-5 digit add-on code is incapable

7.2.5 UPC-A

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



**UPC-A check digit output is allowed



Prohibits UPC-A check bit output

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the UPC-A barcode reading function.



**Allow reading UPC-A barcode



UPC-A literacy is prohibited

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or disallow the forced output of UPC-A additional codes.



UPC-A mandatory output of additional codes



**UPC-A does not require the output of additional codes

Users can read the following setting code to enable and prohibit configurations related to UPCA additional codes.



UPC-A-2-bit add-on code enablement



**UPC-A-2-digit appendix code is disabled



UPC-A-5-bit add-on code enablement



**UPC-A-5-digit add-on code is disabled

Users can read the following setting code to configure UPCA leading characters and system characters.



UPCA Transport System Characters



Do not transmit leading characters (do not transmit country codes and system characters).



Transmit country code + system characters

UPC-A converts EAN13 enable

Users can allow/disable UPCA conversion to EAN13 by reading the following setting codes



Allow UPC-A conversion to EAN13



**Disable UPC-A conversion to EAN13

7.2.6 UPC-E0

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



**UPC-E0 check bit output is allowed



The UPC-E0 check bit output is prohibited

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable UPCA barcode reading functions.



**Allow reading UPC-E0 barcode



Prohibition of UPC-E0 literacy

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or prohibit the forced output function of UPCE0 additional code.



UPC-E0 mandatory output of additional codes



**UPC-E0 does not require an additional code to be output

Users can read the following settings to enable and prohibit UPCE0 additional codes



UPC-E0-2-bit add-on code enablement



**UPC-E0-2-digit add-on code is disabled



UPC-E0-5-digit add-on code enablement



**UPC-E0-5-digit additional code is disabled

Users can read the following setting code to configure UPCA leading characters and system characters.



UPCA Transport System Characters



Do not transmit leading characters (do not transmit country codes and system characters).



Transmit country code + system characters

7.2.7 UPC-E1

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



**UPC-E1 check bit output is allowed



The UPC-E1 check bit output is prohibited

Permission and prohibition of reading function

Users can allow or disable UPCE1 barcode reading function by reading the following setting codes.



**Allow reading UPC-E1 barcode



Prohibition of UPC-E1 literacy

Additional code forced output function allows and prohibits

Users can read the following setting codes to allow or prohibit the forced output function of UPCE1 additional code.



UPC-E1 mandatory output of additional codes



**UPC-E1 does not require additional code output

Users can configure UPCE1 additional code enablement and prohibition by reading the following setting codes.



UPC-E1-2-bit add-on code enablement



**UPC-E1-2-digit add-on code is inhibited



UPC-E1-5-digit add-on code enablement



**UPC-E1-5-digit add-on code is disabled

7.2.8 Code128

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the Code128 barcode reading function.



**Allow reading Code128 barcode



Code128 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Code128 by reading the following setting codes.



Code128 information has a minimum length of 0



**Code128 information has a minimum length of 4



**Code128 information has a maximum length of 32



The maximum length of Code128 information is 255

7.2.9 Code39

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allow Code39 check bit output



**Code39 check digit output is prohibited

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



Code39 validation enablement



**Code39 Proofreading Ability

Permission and prohibition of reading function

Users can allow or disable the Code39 barcode reading function by reading the following setting codes.



**Allow reading Code39 barcode



Code39 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Code39 by reading the following setting codes



The minimum length of Code39 messages is 0



**Code39 information has a minimum length of 4



**Code39 information has a maximum length of 32



The maximum length of a Code39 message is 255

Start and end character output settings

Users can read the following setting code to set the output of Code39 starter and end characters.



Code39 initiation and ender output



**Code39 start and end characters are not output

Code32 pattern

Users can set whether Code39 supports Code32 mode by reading the following setting code.



Supports Code32 mode



*Code32 mode is not supported

Users can set the prefix output of Code32 mode through the following setting code



Code32 prefix output



Code32 prefix is not output

FullAsc mode

Users can set whether Code39 supports FullAsc mode by reading the following setting code.



**Supports FullAsc mode



FullAsc mode is not supported

7.2.10 Code93

Permission and prohibition of reading function

Users can allow or disable the Code93 barcode reading function by reading the following setting codes.



****Allow reading Code93 barcode**



Code93 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Code93 by reading the following setting codes



The minimum length of Code93 information is 0



****Code93 information has a minimum length of 4**



****Code93 messages have a maximum length of 32**



The maximum length of Code93 information is 255

7.2.11 Interleaved 2 of 5

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Enable Interleaved 2 of 5 Check Digit Output



**** Disabled Interleaved 2 of 5 Check Digit Output**

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



Interleaved 2 of 5 Check Digit Enable



****Interleaved 2 of 5 Check Digit Disable**

Permission and prohibition of reading function

Users can allow or disable Interleaved 2 of 5 barcode reading by reading the following setting codes.



Allow reading Interleaved 2 of 5 code



**Interleaved 2 of 5 literacy is prohibited

Readable length setting

Users can set the minimum and maximum read lengths of Interleaved 2 of 5 by reading the following setting codes



Interleaved 2 of 5 messages have a minimum length of 0



**Interleaved 2 of 5 messages have a minimum length of 4



**Interleaved 2 of 5 information has a maximum length of 32



The maximum length of Interleaved 2 of 5 information is 255

Brazilian bank code conversion settings

Users can convert the Interleaved 2 of 5 to Banco do Brasil code by reading the following setting code



Brazilian bank code conversion enabled



** Brazilian bank code conversion ban

7.2.12 Industrial 25

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Enable Industrial 25 Check Digit Output



**Industrial 25 check bit output is prohibited

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



Industrial 25 Check Digit Enable



**Industrial 25 Check Digit Disable

Permission and prohibition of reading function

Users can allow or disable the Industrial 25 barcode reading function by reading the following setting codes.



Allow reading Industrial 25 code



**Industrial 25 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Industrial 25 by reading the following setting codes



The minimum length of the Industrial 25 information is

0



**Industrial 25 information has a minimum length of 4



**Industrial 25 information has a maximum length of 32



The maximum length of the Industrial 25 information is

255

7.2.13 Matrix 2 of 5

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allows Matrix 2 of 5 check bit output



**Prohibits Matrix 2 of 5 check bit output

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



Matrix 2 of 5 Check Digit Enable



**Matrix 2 of 5 Check Digit Disable

Permission and prohibition of reading function

Users can allow or disable the Matrix 2 of 5 barcode reading function by reading the following setting codes.



Allow reading Matrix 2 of 5 code



**Matrix 2 of 5 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Matrix 2 of 5 by reading the following setting codes



The Matrix 2 of 5 message has a minimum length of 0



**Matrix 2 of 5 information has a minimum length of 4



**Matrix 2 of 5 information has a maximum length of 32



The maximum length of the Matrix 2 of 5 message is
255

7.2.14 Code11

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allow Code11 check bit output



**Code11 check bit output is prohibited

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



**Code11 verification enablement



Code11 Verify the inhibition

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the Code11 barcode reading function.



Allow Code11 to read



**Code11 literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of Code11 by reading the following setting codes



The minimum length of Code11 information is 0



**Code11 messages have a minimum length of 4



**Code11 messages have a maximum length of 32



The maximum length of a Code11 message is 255

7.2.15 CodaBar

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the CodaBar barcode reading function.



**Allow reading CodaBar code



CodaBar literacy is prohibited

Readable length setting

Users can set the minimum and maximum reading lengths of CodaBar by reading the following setting codes



The minimum length of the CodaBar information is 0



**CodaBar information has a minimum length of 4



**CodaBar information has a maximum length of 32



The maximum length of the CodaBar information is 255

Verify formatting

Users can set the CodaBar verification format by reading the following setting code



CodaBar Mod16 Check Digit



**Disable CodaBar Check Digit

Start and stop output settings

Users can set the output of the CodaBar start and stop by reading the following setting code



Output CodaBar Start/Stop Characters



**CodaBar start and stop symbols are not output

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allows CodaBar check bit output



**Disable CodaBar check bit output

7.2.16 MSI

Permission and prohibition of reading function

Users can allow or disable the MSI barcode reading function by reading the following setting codes.



Allow MSI to read



** Disable MSI reading

Readable length setting

Users can set the minimum and maximum read lengths of MSI by reading the following setting codes



The minimum length of MSI information is 0



**MSI information has a minimum length of 4



**MSI information has a maximum length of 32



The maximum length of MSI information is 255

Verify formatting

Users can set the MSI verification format by reading the following setting code



**Enable MSI Mod10 Check Digit



MSI Dual Mod10 Calibration

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allows MSI check bit output



**MSI check bit output is prohibited

7.2.17 RSS-14

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the RSS-14 barcode reading function.



RSS-14 literacy is allowed



**禁止 RSS-14 识读

7.2.18 RSS- Ltd

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the RSS-Ltd barcode reading function.



Allow RSS-Ltd to read



**Disable RSS-Ltd Reading

7.2.19 RSS- Exp

Permission and prohibition of reading function

Users can read the following setting codes to allow or disable the RSS-Exp barcode reading function.



Allow RSS-Exp to read



**Disable RSS-Exp Reading

Readable length setting

Users can set the minimum and maximum reading lengths of RSS-Exp by reading the following setting codes



The minimum length of RSS-Exp messages is 0



**RSS -Exp The minimum length of the message is 4



**RSS-Exp messages have a maximum length of 32



The maximum length of RSS-Exp messages is 255

7.2.20 Plessey

Permission and prohibition of reading function

Users can allow or disable the Plessey barcode reading function by reading the following setting codes.



Allow Plessey to read



**Disable Plessey Reading

Readable length setting

Users can set the minimum and maximum reading lengths of Plessey by reading the following setting codes



The shortest length of the Plessey message is 0



**Plessey information has a minimum length of 4



**Plessey messages have a maximum length of 32



The maximum length of the Plessey message is 255

Validation enabled settings

Users can set up product barcode verification by reading the following setting codes



Enable Plessey Check Digit Verification



** Disable Plessey Check Digit Verification

Check bit output settings

Users can allow/disallow the output of product barcode checkpoints by reading the following setting codes



Allows Plessey check bit output



*Prohibits Plessey check bit output

7.3 QR code operation

7.3.1 QR Code

Users can read the following setting codes to allow or disable the QR Code reading function.



**QR literacy is allowed



QR literacy is prohibited

7.3.2 Data Matrix(DM)

You can read the following settings to allow or disable the Data Matrix reading function.



**Allow reading DM code



DM literacy is prohibited

7.3.3 PDF417

Users can read the following setting code to allow or disable PDF417 reading functions.



**Allow reading PDF417 code



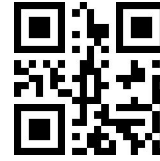
PDF417 literacy is prohibited

7.3.4 Micro PDF417

Users can read the following setting code to allow or disable the Micro PDF417 reading function.



****Allow reading Micro PDF417 code**



Micro PDF417 literacy is prohibited

7.3.5 Micro QRCode

Users can read the following setting code to allow or disable the Micro QRCode reading function.



Allow reading Micro QRCode code



****Micro QRCode literacy is prohibited**

7.3.6 MaxiCode

Users can allow or disable MaxiCode reading functions by reading the following setting codes.



Allow reading MaxiCode code



****MaxiCode literacy is prohibited**

7.3.7 Aztec Code

Users can allow or disable the Aztec Code reading function by reading the following setting codes.



Allow reading Aztec Code code



****Aztec Code literacy is prohibited**

8 Save and cancel

After reading the Data Code, you need to read the Save Data setting code before you can save the data. When there is an error in reading data, the "Cancel Data" setting code can be read to cancel.



Data preservation



Data cancellation

appendix

Appendix A: Data Code

0~9



0



1



2



3



4



5



6



7



8



9

A~F



A



B



C



D



And



F

Appendix B: Default Settings Table

Parameter Name		Default settings	remark
Set code			
Set code function		Open	
Communication settings			
Communication mode		USB-HID	
TTL-232	Serial port baud rate	9600bps	
	Serial check bit	There is no check position	
USB-HID	HID data sending interval	1ms	Range: 1ms~16ms
	CapsLock settings	No conversion	
Scan code mode parameters			
Silent reading mode		Button trigger mode	
Triggered by the key	Trigger method	Edge trigger	
	Single reading time	infinite length	Range: 1~infinite length
Continuous code reading	Single reading time	5s	Range: 1~infinite length
	The length of the reading interval	1s	Range: 0~5s
	The same code reading delay	Shut down	
Automatic sensing	Single reading time	5s	Range: 1~infinite length
	The length of the reading interval	1s	Range: 0-5s
	The same code reading delay	Shut down	
	sensitivity	Medium sensitivity	
General settings			
Fill light and positioning	Fill light	Light up when taking photos	
	Positioning light	Light up when taking photos	
buzzer	mute	Shut down	
	Power on prompt	Open it	
	Decode the prompt	Open it	
	Set code prompt	Open it	
	Decode the prompt duration	60ms	Range: 30~120ms

	Buzzer settings	Passive buzzer	
	Passive buzzer	Medium frequency sound	
	Active buzzer	High level	Low levels are effective when idle and high levels are effective when working
Output data format		UTF-8	
Virtual keyboard enablement		standard	
Control how characters are output		Ctrl mode	
Keyboard language		United States	
Data editing			
prefix		not	
suffix		not	
CODE ID		Shut down	
Terminator		enter	
Data segment interception		Output full data segments	
Code settings			
Reading type		Silently recognize the type of reading	
EAN13			
Check bit output		forbid	
Literate		allow	
Force output of additional codes		Not required	
2-digit additional code		forbid	
5-digit additional code		forbid	
EAN8			
Check bit output		allow	
Literate		allow	
Force output of additional codes		Not required	
2-digit additional code		forbid	
5-digit additional code		forbid	
UPC-A			
Check bit output		allow	
Literate		allow	
Force output of additional codes		Not required	
2-digit additional code		forbid	
5-digit additional code		forbid	
PUC-A turn EAN13		forbid	
UPC-E0			
Check bit output		allow	
Literate		allow	
Force output of additional codes		Not required	

2-digit additional code	forbid	
5-digit additional code	forbid	
UPC-E1		
Check bit output	allow	
Literate	allow	
Force output of additional codes	Not required	
2-digit additional code	forbid	
5-digit additional code	forbid	
Code128		
Literate	allow	
The shortest length of reading	4	
The longest length of reading	32	
Code39		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	allow	
The shortest length of reading	4	
The longest length of reading	32	
Start and stop symbols	No output	
Code 32 mode	Not supported	
FullAsc mode	In the tank	
Code93		
Literate	allow	
The shortest length of reading	4	
The longest length of reading	32	
Interleaved 2 2 of 5		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
Industrial 25		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
Matrix 2 of 5		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	

Code11		
Check bit output	forbid	
Verify and enable	Enable	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
CodaBar		
Literate	allow	
The shortest length of reading	4	
The longest length of reading	32	
Verify the format	Shut down	
Start and stop symbols	No output	
Check bit output	forbid	
MSI		
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
Verify the format	MSI Mod10 calibration	
Check bit output	forbid	
RSS-14		
Literate	forbid	
AI format	RSS-14 AI without parentheses	
Check bit output	forbid	
RSS- Ltd		
Literate	forbid	
AI format	RSS-Ltd AI without parentheses	
Check bit output	forbid	
RSS- Exp		
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
AI format	RSS-Exp AI without parentheses	
Check bit output	forbid	
Standard 2 of 5		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	

Plessey		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
China Post 25		
Check bit output	forbid	
Verify and enable	Forbidden energy	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
Code16K		
Check bit output	forbid	
Literate	forbid	
The shortest length of reading	4	
The longest length of reading	32	
GS1 Composite		
Literate	allow	
QR Code		
Literate	allow	
QR Code ECI output	No output	
Data Matrix		
Literate	allow	
Multi-code reading	allow	
DM ECI output	No output	
PDF417		
Literate	allow	
Han Xin		
Literate	forbid	
Han Xin ECI output	No output	
Micro PDF417		
Literate	forbid	
Micro QRCode		
Literate	forbid	
MaxiCode		
Literate	forbid	
MaxiCode ECI output	No output	
Aztec Code		
Literate	forbid	
Aztec Code ECI output	No output	

Appendix C: Code ID List

Barcode type	Corresponding characters	Logo bit address
EAN-13	d	0x91
EAN-8	d	0x92
UPC-A	c	0x93
UPC-E0	c	0x94
UPC-E1	c	0x95
Code 128	j	0x96
Code 39	b	0x97
Code 93	i	0x98
Codabar	a	0x99
Interleaved 2 of 5	and	0x9A
Industrial 2 of 5	D	0x9B
Matrix 2 of 5	v	0x9C
Code 11	H	0x9D
MSI-Plessey	m	0x9E
GS1 Databar(RSS-14)	R	0x9F
GS1 Databar Limited(RSS)	R	0xA0
GS1 Databar Expanded(RSS)	R	0xA1
Standard 2 of 5	s	0xA5
Plessey	P	0xA6
ChinaPost 25	p	0xA7
Code16K	k	0xA8
GS1 Composite	G	0xA9
QR Code	Q	0xA2
Data Matrix	in the	0xA3
PDF 417	r	0xA4
Han Xin	X	0xAA
Micro PDF417	r	0xAB
Micro QRCode	Q	0xAC
MaxiCode	M	0xAD
Aztec Code	A	0xAE

Appendix D: ASCII Code Table

hexadecimal	decimal	character
00	0	NOBODY
01	1	SOH
02	2	STX
03	3	ETX
04	4	EOT
05	5	ENQ
06	6	ACK
07	7	BEL
08	8	BS
09	9	HT
0a	10	LF
0b	11	VT
0c	12	FF
0d	13	CR
0e	14	SO
0f	15	YES
10	16	According to
11	17	DC1
12	18	DC2
13	19	DC3
14	20	DC4
15	21	TO
16	22	AMAZED
17	23	ETB
18	24	CAN
19	25	IN
1a	26	SUB
1b	27	ESC

1c	28	FS
1d	29	GS
1e	30	RS
1f	31	US
20	32	SP
21	33	!
22	34	"
23	35	#
24	36	\$
25	37	%
26	38	&
27	39	`
28	40	(
29	41	)
2a	42	*
2b	43	+
2c	44	,
2d	45	-
2e	46	.
2f	47	/
30	48	0
31	49	1
32	50	2
33	51	3
34	52	4
35	53	5
36	54	6
37	55	7
38	56	8
39	57	9
3a	58	:
3b	59	;
3c	60	<
3d	61	=
3e	62	>

3f	63	?
40	64	@
41	65	A
42	66	B
43	67	C
44	68	D
45	69	And
46	70	F
47	71	G
48	72	H
49	73	I
4a	74	J
4b	75	K
4c	76	L
4d	77	M
4e	78	N
4f	79	Or
50	80	P
51	81	Q
52	82	R
53	83	S
54	84	T
55	85	In the
56	86	V
57	87	In
58	88	X
59	89	And
5a	90	With
5b	91	[
5c	92	\
5d	93	]
5e	94	^
5f	95	—
60	96	'
61	97	a

62	98	b
63	99	c
64	100	d
65	101	and
66	102	f
67	103	g
68	104	h
69	105	i
6a	106	j
6b	107	k
6c	108	l
6d	109	m
6e	110	n
6f	111	or
70	112	p
71	113	q
72	114	r
73	115	s
74	116	t
75	117	in the
76	118	v
77	119	in
78	120	x
79	121	and
7a	122	with
7b	123	{
7c	124	
7d	125	}
7e	126	~
7f	127	Of the

Appendix E: List of AIM IDs

Barcode type	AIM ID	Possible AIM ID parameters (m)
Aztec	]z0	0
Codabar	]Fm	0,2,4
Code 39	]On	0, 1, 3, 4, 5, 7
Code 93	]G0	0
Code 128	]C0	0
GS1 DataBar(RSS14)	]e0	0
GS1 DataBar Expanded	]e0	0
GS1 DataBar Limited	]e0	0
Data Matrix	]d1	1
EAN-8	]ln	3, 4
EAN-13	]ln	0, 3
UPC-A	]ln	0, 3
UPC-E	]ln	0, 3
Interleaved 2 of 5	]ln the	0, 1, 3
Industrial 2 of 5	]S0	0
Matrix 2 of 5	]X0	0
PDF417	]L2	2
Micro PDF417	]L2	2
QR	]Q1	1
Micro QR	]Q1	1
MSI Plessey	]Mm	0, 1
Code 11	]Hm	0, 1, 3
Code 32	]X0	0
ISSN	]X5	5
ISBN	]X4	4
Maxicode	]ln order to	0, 1
China Post 25	]X0	0

Reference: **ISO/IEC 15424-2008** Information technology - Automatic identification and data acquisition techniques - Data carrier identifiers (including symbolic identifiers).

Appendix F: Common Serial Commands

function	Serial port command
Set the baud rate to 9600	7E 00 08 02 00 2A 39 01 A7 EA
Save settings to internal Flash	7E 00 09 01 00 00 00 DE C8

Query baud rate	7E 00 07 01 00 2A 02 D8 0F
-----------------	----------------------------

After the host sends a serial command to query the baud rate, the reading engine replies with the following message:

The corresponding baud rate	Back to info
1200	02 00 00 02 C4 09 SS SS
4800	02 00 00 02 71 02 SS SS
9600	02 00 00 02 39 01 SS SS
14400	02 00 00 02 D0 00 SS SS
19200	02 00 00 02 9C 00 SS SS
38400	02 00 00 02 4E 00 SS SS
57600	02 00 00 02 34 00 SS SS
115200	02 00 00 02 1A 00 SS SS

Note: SS SS is the checkcase.